

Homework 9 - Results

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Running Task 1: CPU Scan
CPU-Avg-Time for 100000 elements: 0.22859573 ms
CPU-Avg-Time for 200000 elements: 0.45950413 ms
CPU-Avg-Time for 300000 elements: 0.70350170 ms
CPU-Avg-Time for 400000 elements: 0.95229149 ms
CPU-Avg-Time for 500000 elements: 1.20651722 ms
CPU-Avg-Time for 600000 elements: 1.50089264 ms
CPU-Avg-Time for 700000 elements: 1.74140930 ms
CPU-Avg-Time for 800000 elements: 1.96731091 ms
CPU-Avg-Time for 900000 elements: 2.31409073 ms
CPU-Avg-Time for 1000000 elements: 2.68909931 ms
Running Task 2: GPU Work-Efficient Scan
Reductions match.
GPU-Avg-Time for 100000 elements: 0.24160695 ms
Reductions match.
GPU-Avg-Time for 200000 elements: 0.17926383 ms
Reductions match.
GPU-Avg-Time for 300000 elements: 0.18276095 ms
Reductions match.
GPU-Avg-Time for 400000 elements: 0.17932415 ms
Reductions match.
GPU-Avg-Time for 500000 elements: 0.18331409 ms
Reductions match.
GPU-Avg-Time for 600000 elements: 0.62073302 ms
Reductions match.
GPU-Avg-Time for 700000 elements: 0.64562798 ms
Reductions match.
GPU-Avg-Time for 800000 elements: 0.63365221 ms
Reductions match.
GPU-Avg-Time for 900000 elements: 0.62658620 ms
Reductions match.
GPU-Avg-Time for 1000000 elements: 0.64174080 ms
```

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num_elements	CPU (ms)	GPU (ms)	Speedup-Factor
100000	0.2286	0.2416	0.95
200000	0.4595	0.1793	2.56
300000	0.7035	0.1828	3.85
400000	0.9523	0.1793	5.31
500000	1.2065	0.1833	6.58

num_elements	CPU (ms)	GPU (ms)	Speedup-Factor
600000	1.5009	0.6207	2.42
700000	1.7414	0.6456	2.70
800000	1.9673	0.6337	3.10
900000	2.3141	0.6266	3.69
1000000	2.6891	0.6417	4.19

Plotted Execution times of GPU and CPU

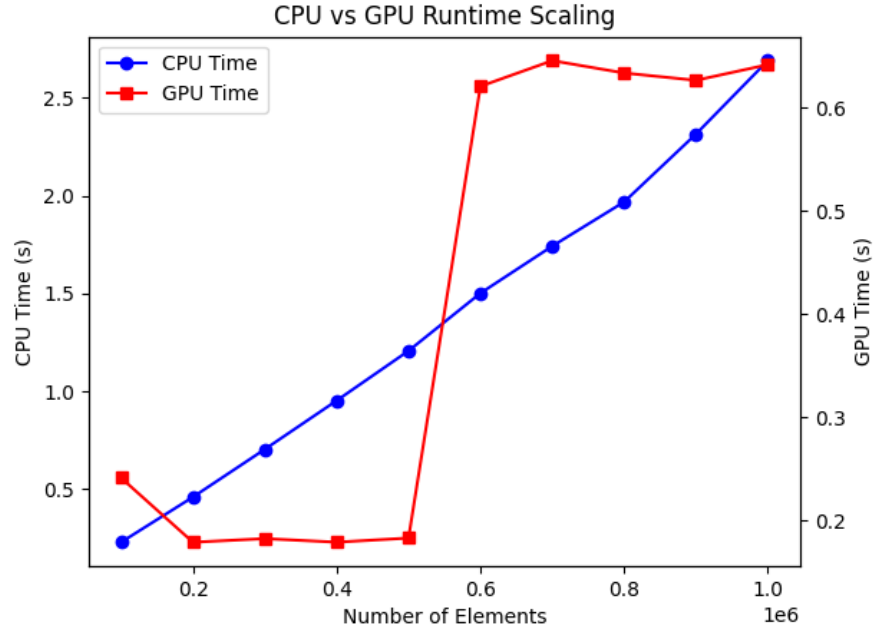


Figure 1: Execution times of the CPU and GPU scan algorithm for different number of elements. (Note: the y-axis are different for CPU and GPU)

Plotted Speedup-Factor

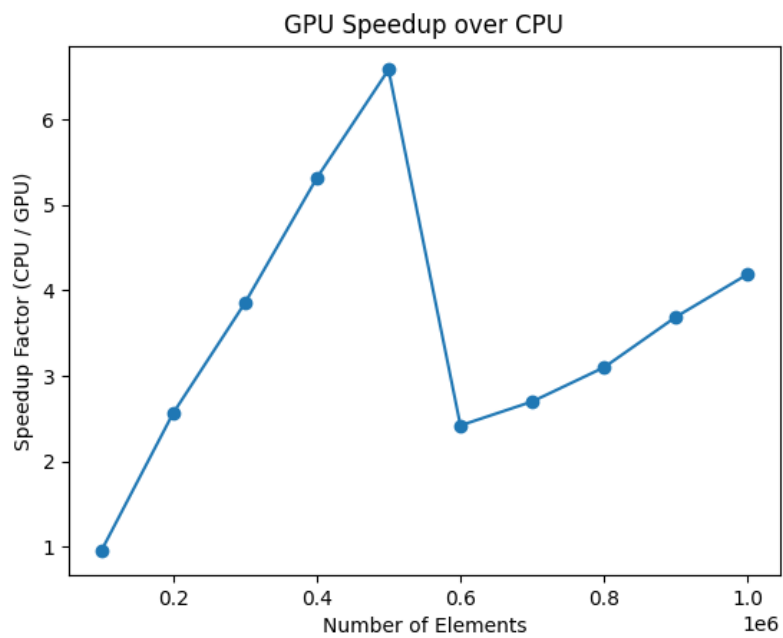


Figure 2: Speedup-Factor CPU/GPU of the scan algorithm for different number of elements